

Harold Le

U.S. Permanent Resident | +1 902-626-5831 | haroldle2735@gmail.com | linkedin.com/in/harold-le | github.com/Hale0123

EDUCATION

McMaster University

Hamilton, ON

Bachelor of Engineering | Software Engineering

Graduation Year: 2027

- Cumulative GPA: 3.5/4.0
- Coursework: Digital Systems, Data Structure and Algorithm, Discrete Math, Intro to Software Development, Software Development Basics, Computer Architecture

EXPERIENCE

Software Quality Assurance Analyst

May 2025 – Sept 2025

QEMS Inc.

Charlotte, NC

- Verify correct deployment of software programs onto industrial control boards and hardware systems
- Conducted functional and integration tests to ensure firmware operated reliably under production conditions
- Logged and documents test cases, anomalies and edge-case behaviour using internal QA tools
- Collaborated with engineers to troubleshoot hardware-software interfacing issues during system bring-up

McMaster Delta Hacks XI

Jan 2025 – Jan 2025

McMaster University

Hamilton, ON

- Participated in a 24-hour hackathon at McMaster University with 600+ participants from across Canada.
- Co-developed **Nutrilens**, a full-stack web app that uses image recognition to analyse food and display nutritional information (calories, macros, allergens, etc.)
- Build the frontend with **ReactJS** and the backend using **Flask**, **Python**, and **Node.js**; integrated the **Nutritionix API** for food data and used **OpenCV** and **TensorFlow** for camera-based food recognition

PROJECTS

Visual Island Scanning and Terrain Algorithm

Feb 2025 – Mar 2025

McMaster University

Hamilton, ON

- Developed a Java-based drone exploration algorithm to auto scan unknown terrain using sensor feedback.
- Implemented fog-of-war visualization and POI detection across various map datasets.
- Collaborated with a team of 3 to design an efficient pathfinding and scanning strategy.
- Utilize Maven, Git, and JSON parsing to integrate with the game engine and simulate output in JSON format.

Personal Portfolio

May 2024 – Present

McMaster University

Hamilton, ON

- Developed a responsive portfolio that showcases personal skills, traits, and experiences.
- Used **ReactJS** framework with **CSS** to build the frontend and **JavaScript** to make the website functional.
- Learned to design layouts and deploy web pages via **GitHub Pages** branch.

Snake And Ladders Game

Nov 2024 – Dec 2024

McMaster University

Hamilton, ON

- Snake And Ladders game developed using **C** and **GTK** for GUI in Linux environment.
- Automate GitHub commit and push using **Bash Scripting**.
- Wrote test cases for bugs test and run GCOV report to evaluate the code coverage.
- Created .YML file for the CI Pipelines to automate the build, test and deployment of code with GitHub Actions.

SKILLS

- Concept & Practices: CI/CD, DevOps, Design Patterns, Algorithms, Version Control, Shell Scripting
- Programming: Python, Java, HTML, CSS, JavaScript, MATLAB, Bash, C, Verilog
- Tools and Framework: ReactJS, JSON, Bootstrap, Vite, Linux, Git, GitHub, TensorFlow, Matplotlib, GTK
- Development Tools: Apache Maven, Vim, VS Code, Jupiter Notebook, PyCharm, PowerShell, Ubuntu, Anaconda